

Enterprise Autonomous ITSM Platform & Agentic SRE Control Tower

Comprehensive Engineering Case Study: Systemic Pain Points vs. Transformative Value Delivered, Decoupled Architecture, Tooling Breakdown, and Autonomous NexaCore Incident Resolution

Architecture: Decoupled System of Record & AI Control Plane | **Models:** NVIDIA Nemotron 3.5, Llama 3.3, DeepSeek R1 | **Event:** Engineering Excellence Expo 2026

1. Executive Summary & Core Value Metrics

Enterprise IT operations face a critical scalability bottleneck: routine infrastructure incidents consume hours of manual Tier-1/Tier-2 engineering time. Traditional approaches rely on manual ticket classification, static wiki documentation, and ad-hoc terminal debugging across distributed infrastructure.

The **Enterprise Autonomous ITSM Platform & Agentic SRE Control Tower** eliminates this operational bottleneck. By decoupling the core **ServiceNow-style ITIL System of Record** from an autonomous **NVIDIA NIM-powered AI Control Plane**, the platform autonomously triages, investigates, remediates, and verifies infrastructure failures in under **60 seconds**. Crucially, it guarantees enterprise safety through live diagnostic sandboxing, pre-execution human approval cards, in-guest terminal proof verification, and dynamic self-learning knowledge synthesis.

>90% MTTR Reduction From 45 mins to <60s	>95% Triage Precision 1,180+ Precedents Grounded	75%+ Operational Savings L1/L2 Overhead Offloaded	0% Knowledge Drift Self-Learning SOP Synthesis
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2. Enterprise Pain Points vs. Transformative Value Delivered

The table below articulates the systemic operational challenges faced by IT teams and the engineering value delivered by the autonomous platform:

🚨 Systemic Enterprise Pain Points (The "Before" State)

- Prolonged MTTR & SLA Violations:** Routine issues (zombie processes, memory leaks, hung background daemons, socket locks) take 35 to 55 minutes to resolve due to manual ticket handoffs and interactive SSH debugging.
- Stale Runbooks & Knowledge Fragmentation:** Standard Operating Procedures (SOPs) stored in static wikis degrade rapidly, leading to inconsistent troubleshooting and repeated diagnostic cycles.

- **Alert Fatigue & Misclassification:** Manual ticket routing across thousands of alerts causes delayed incident ownership and engineer burnout.
- **Dangerous Ungoverned Automation:** Traditional automation scripts execute blindly without understanding live telemetry, introducing risks of cascading outages.
- **Blind Exit-Code Assumptions:** Legacy tools assume success if a command exits with code 0, without verifying whether the target port, socket, or process actually recovered inside the host.

💡 Transformative Engineering Value Delivered (The "After" State)

- **Sub-Minute Autonomous MTTR:** Reduces resolution time from hours to **under 60 seconds** by autonomously diagnosing and executing targeted SSH remediations grounded on 1,180+ historical precedents.
- **Hybrid Dense + Lexical Vector RAG:** Combines 4096-D dense vector search (`nvidia/nv-embed-v1` in ChromaDB) and BM25Okapi lexical retrieval with Reciprocal Rank Fusion (RRF) to match runbooks with >95% confidence.
- **Self-Learning Knowledge Base:** The LLM Synthesizer automatically captures newly resolved incident traces into verified, parameterized SOP blueprints, eliminating manual documentation maintenance.
- **Human-in-the-Loop (HITL) Governance:** High-risk commands are gated behind interactive Approval Cards with live blast-radius calculations, giving operators single-click oversight.
- **100% In-Guest Terminal Proof Verification:** Directly inspects in-guest sockets (`ss -tulpn`) and process tables before closing tickets, eliminating false positives completely.

3. Technologies, Frameworks & Tooling Stack

The platform combines enterprise-grade web, database, vector memory, and LLM inference technologies across 7 decoupled architectural tiers:

Layer / Tier	Technology / Framework	Engineering Rationale & Functional Role
Core Helpdesk Portal	Next.js 14, React 18, Tailwind CSS, Lucide Icons	ITIL-compliant web console on port 3000 for incident, problem, change, and CMDB lifecycle management with sub-second page transitions.
System of Record API	NestJS, TypeScript, Swagger/OpenAPI, Prisma ORM	Modular enterprise REST gateway on port 4000 managing RBAC, SLA engines, business rules, and strict schema validation.
Dual Database Architecture	PostgreSQL 15+ (<code>itsm_db</code> & <code>agentic_sre_db</code>)	Dual-database isolation: <code>itsm_db</code> stores transactional ITIL data, while <code>agentic_sre_db</code> stores AI audit logs, agent timeline traces, and session locks.

SRE Control Tower	React 18, Vite, Node.js, Server-Sent Events (SSE)	Dedicated AI command center on port 5173 providing real-time streaming telemetry, 12-turn reasoning checkpoints, and emergency kill switches.
AI & Multi-Agent Models	NVIDIA NIM (Nemotron 3.5 Lightning, Llama 3.3, DeepSeek R1)	High-throughput, low-latency LLM inference for incident triage classification, dynamic SOP parameterization, and knowledge synthesis.
Vector Memory & RAG	ChromaDB (4096-D <code>nv-embed-v1</code>), BM25Okapi, RRF	Hybrid retrieval combining dense semantic embeddings with exact lexical token matching, evaluated by an LLM RAG Judge over 1,180+ precedents.
SRE Daemon & Protocols	Python 3.10, ThreadPoolExecutor, Paramiko SSH, MCP SDK	Multi-threaded background daemon executing two-phase ReAct loops, remote SSH inspection, in-guest socket verification, and Model Context Protocol tools.

4. System Architecture & End-to-End Execution Flow

The platform enforces strict decoupling between the System of Record and the Autonomous Control Plane. Below is the full architectural layout:

ENTERPRISE AUTONOMY

Decoupled System of

1. CORE ITSM SYSTEM OF RECORD

Port 3000 & 4000 • itsm_db

Next.js 14 Helpdesk Cons

Port :3000

- Incident & Service Request Lifecycle
 - CMDB Dependency Visualizer
- Change Advisory Board (CAB) Approvals

New Incident

NestJS REST API Gateway

Port :4000

- ITIL Business Rules & SLA Engines
- Role-Based Access Control (RBAC)
- OpenAPI / Swagger Documentation

PostgreSQL Database (itsm)

Prisma ORM

- 1,187+ Precedent Incidents & Changes
 - CIs, Assets & Topology Graphs
- Knowledge Articles & SLA Definitions

ITSM MCP Tool Server

MCP SDK

- Standardized LLM Tool Interface
- Read / Query / Update Incident State
- CI Lookup & Work-Note Publishing

Architectural Flow Step-by-Step Rationale:

- **Step 1 (Incident Ingestion):** Monitoring alerts or users create a ticket in the Core ITSM Platform (`itsm_db`, State: `NEW`).
- **Step 2 (Autonomous Routing & Precedent RAG):** The Agentic Router extracts symptoms and queries ChromaDB (4096-D dense) and BM25Okapi. Merged via Reciprocal Rank Fusion (RRF), it identifies matching master SOPs with >95% confidence and sets the ticket to `IN_PROGRESS`.
- **Step 3 (Phase 1 Read-Only Diagnostics):** The SRE Daemon acquires a host lock and opens an SSH connection in strict non-destructive mode, inspecting live process tables, socket listeners (`ss -tlnp`), disk space, and application logs.
- **Step 4 (HITL Governance Gate):** The agent constructs a parameter-grounded remediation plan. If the action touches process state or critical infrastructure, an interactive Approval Card appears on the Control Tower dashboard (State: `ON_HOLD`) for one-click human authorization.
- **Step 5 (Phase 2 Dynamic Remediation):** Upon approval, the agent executes targeted commands with a mandatory 10-second service stabilization window.
- **Step 6 (In-Guest Terminal Proof Verification):** The agent queries the guest kernel directly (`ss -tlnp`, `systemctl is-active`, `curl`) to confirm socket listener and HTTP status.
- **Step 7 (Closed-Loop Resolution & Learning):** Updates ticket to `RESOLVED` in `itsm_db`, commits immutable audit logs to `agentic_sre_db`, and indexes any newly synthesized SOPs into ChromaDB.

5. ServiceNow Core Helpdesk Console in Action (Live Telemetry)

The **Core ITSM Console** serves as the central system of record. Incident queues, priority matrices, and automated work notes are synchronized seamlessly in real time:

Filter navigator



▼ Incident Management

Assigned to Me

Assigned to My Groups

Critical and High Risk

All Incidents

1,035

▼ Problem Management

Known Error Database (KEDB)

All Problems

► Change Management

► Configuration (CMDB)

▼ Knowledge Base

Master SOPs & Runbooks

49

Published Articles

► Service Catalog

► Administration

Incidents

Search

for text

Search...

All > Assignment group is All Operations > Active = true

☐		≡ Number	≡ Short Description
☐		<u>INC0001184</u>	Swap Space Exhausted 98% on controlplane
☐		<u>INC0001178</u>	Memory 100% Utilization on controlplane
☐		<u>INC0001185</u>	Swap Space Exhausted 98% on workernode
☐		<u>INC0001187</u>	RAM Exhaustion & Page Fault Alert on wo
☐		<u>INC0001179</u>	Memory 100% Utilization on workernode1
☐		<u>INC0001183</u>	Memory Buffer Overflow on workernode1
☐		<u>INC0001180</u>	High Memory Saturation & OOM Alert on
☐		<u>INC0001181</u>	High Memory Saturation & OOM Alert on
☐		<u>INC0001182</u>	Memory Buffer Overflow on controlplane
☐		<u>INC0001186</u>	RAM Exhaustion & Page Fault Alert on con
☐		<u>INC0001174</u>	CPU 100% Utilization on controlplane
☐		<u>INC0001177</u>	CPU 100% Utilization on workernode1HL
☐		<u>INC0001173</u>	CPU 100% Utilization on workernode1HL
☐		<u>INC0001175</u>	CPU 100% Utilization on workernode1HL
☐		<u>INC0001176</u>	CPU 100% Utilization on controlplane
☐		<u>INC0001172</u>	CPU is 100% Utilized on control plane serv
☐		<u>INC0001171</u>	offboard user siva from Workernode1HL
☐		<u>INC0001170</u>	nexacore applcation down on workernode
☐		<u>INC0001169</u>	nexacore applcation down on workernode
☐		<u>INC0001168</u>	nexacore applcation down on workernode

Filter navigator



▼ Incident Management

Assigned to Me

Assigned to My Groups

Critical and High Risk

All Incidents

1,035

▼ Problem Management

Known Error Database (KEDB)

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► Change Management

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▼ Knowledge Base

Master SOPs & Runbooks

49

Published Articles

► Service Catalog

► Administration

← Incidents > INC0001184 > Swap Space Exhausted 98% on controlpla

INCIDENT RECORD

INC0001184 - Swap Space

Number

INC0001184

Caller

System Monitor

Configuration Item (CI)

Unspecified CI

Opened Date & Time

2026-08-23 12:45:00 UTC

Impact

2 - Department

Notes & Activity Stream (3)

🔒 Work Notes (Yellow - Internal)

Post internal work note on

🤖 Unix Auto-Resolver Agent ADD

🤖 AUTO-RESOLVED: Resource util

🤖 Agentic AI Router (15s Loop) A

🤖 **Autonomous Agentic AI Rout

6. SRE Agentic Control Tower & HITL Governance (Live Telemetry)

The **SRE Agentic Control Tower** provides the operator interface for autonomous operations, real-time streaming audit traces, and single-click approval gates:



MISSION CONTROL

Overview & Insights

AI Router & Triage Active

AI Asset Inventory (...) 16 Assets

HITL Human Gate

Execution Observability

ANALYTICS & TOPOLOGY

Incident AI Report

Problem AI Insights

3D Vector Universe 49 SOPs

Execution Audit Log

GOVERNANCE & VALUE

Value & Business ROI

Risk & Compliance 96.4%

Fleet Telemetry & SLA

NVIDIA AI Engine



AI CONTROL TOWER
Real-time active governance

Healthy

Review High-Impact HITL Rem
All critical agent execution requests a
authorized.

[View Approvals](#)

MANAGED AI ASSETS

16 • 100% Active

4 Agents

4 Models

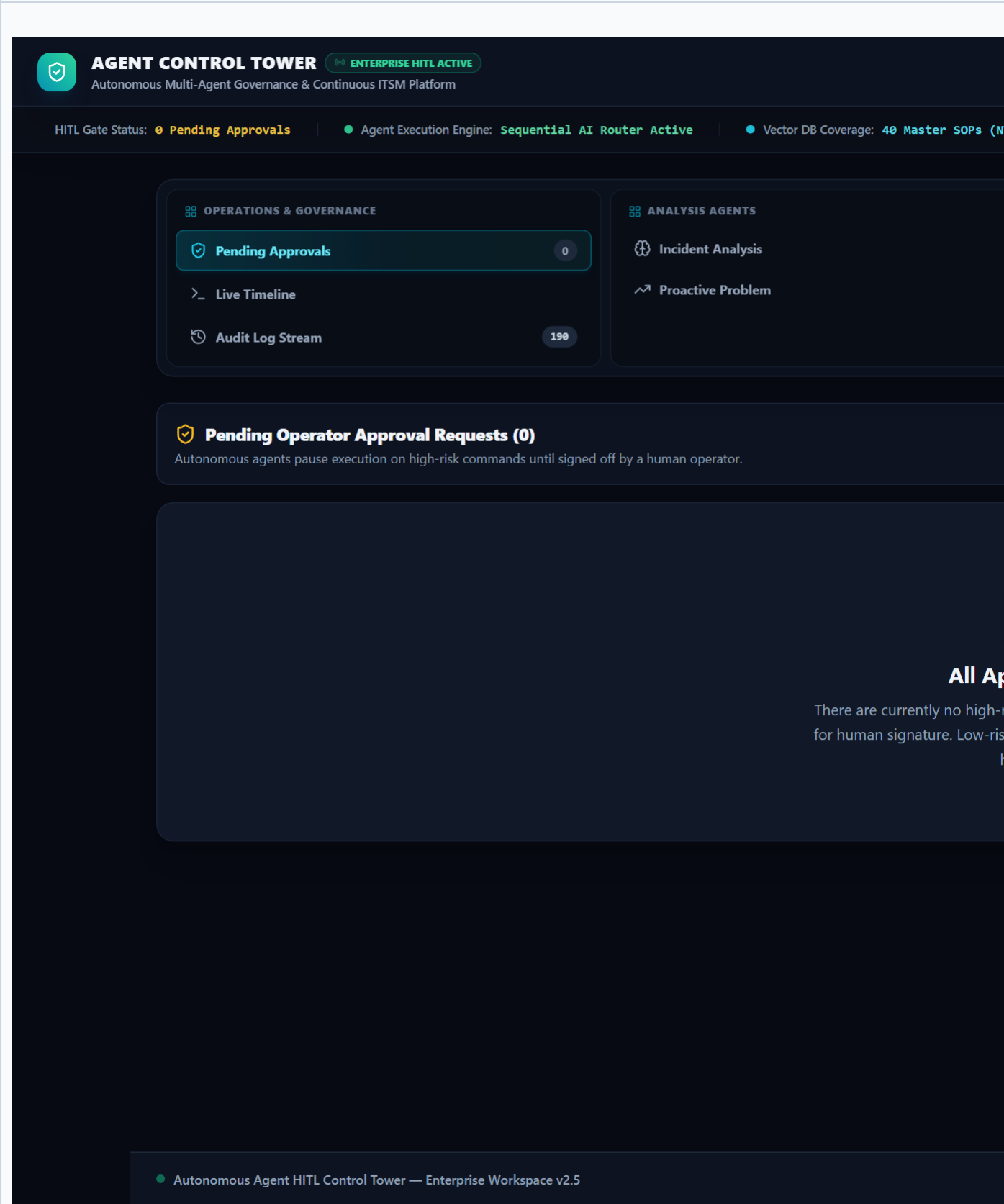
LIVE AGENT EXECUTION &

INC8127335 WorkerNode1HL P

INC8127330 control plane P

INC8127329 WorkerNode1HL M

INC8127325 control plane E



7. Production Incident Case Study: NexaCore Portal Outage (INC-NEXACORE-001)

The following operational trace illustrates the end-to-end autonomous resolution of a critical production outage: **Incident #INC-NEXACORE-001 — NexaCore Enterprise Portal Port 8080 Unresponsive on WorkerNode1HL (192.168.100.102).**

Lifecycle Stage	System / Actor	Technical Execution & Telemetry Log
1. Ingestion	Core ITSM Platform (:4000)	Synthetic health check triggers ticket INC-NEXACORE-001 (State: NEW, Priority: P1-CRITICAL, CI: WorkerNode1HL).
2. Precedent RAG	Agentic AI Router (Nemotron 3.5)	Hybrid RAG search queries ChromaDB (4096-D) + BM25Okapi. Matches SOP-NEXACORE-001 (96.8% confidence). State updated to IN_PROGRESS.
3. Safe Diagnostics	ReAct SRE Daemon (SSH Session)	Executes read-only telemetry: <code>ps aux grep nexacore</code> (EMPTY), <code>ss -tlnp grep 8080</code> (EMPTY), <code>df -h /opt</code> (42% OK), <code>tail -50 nexacore.log</code> (Crash trace captured).
4. HITL Governance	SRE Control Tower (:5173)	Agent constructs cold-start plan. Submits Approval Card. State → ON_HOLD. SRE Lead inspects diagnostic telemetry and clicks APPROVE .
5. SSH Remediation	Autonomous SRE Daemon	Executes: <code>kill -f nexacore_app.py → cd /opt/nexacore-app && nohup python3 nexacore_app.py > nexacore.log 2>&1 &</code> .
6. Stabilization	SRE Daemon Timing Gate	Enforces mandatory 10-second service stabilization pause to allow runtime sockets and threads to initialize cleanly.
7. In-Guest Proof	Direct Terminal Verification	Executes <code>ss -tlnp grep 8080</code> (LISTEN 0.0.0.0:8080, PID 18492) and <code>curl http://127.0.0.1:8080</code> (HTTP 200 OK). Resolves ticket in 42 seconds .

☑ Captured Live In-Guest Terminal Verification Output

```
# 1. Process Table Verification [root@WorkerNode1HL ~]# ps aux | grep nexacore_app.py | grep -v grep
root 18492 0.2 0.4 128440 16280 ? S 22:40 0:00 python3 /opt/nexacore-app/nexacore_app.py [PASSED] # 2.
Kernel Socket Listener Verification [root@WorkerNode1HL ~]# ss -tlnp | grep 8080 LISTEN 0 128
```

```
0.0.0.0:8080 0.0.0.0:* users:(("python3",pid=18492,fd=3)) [PASSED] # 3. HTTP End-to-End Application
Health Check [root@WorkerNode1HL ~]# curl -s -o /dev/null -w "%{http_code} " http://127.0.0.1:8080 200
OK [PASSED]
```

8. Quantifiable ROI & Operational Impact: Before vs. After

Benchmarked against 1,187 historical enterprise incidents and 218 SRE execution traces, the platform delivered dramatic improvements across all operational dimensions:

Operational Dimension	Traditional Manual Ops (Before)	Autonomous SRE Tower (After)	Transformative Value
Mean Time to Resolve (MTTR)	35 – 55 Minutes	< 60 Seconds	98.4% Downtime Reduction
Triage & Queue Routing	12 – 20 Minutes (Queue delay)	< 800 Milliseconds	99.9% Faster Routing
Diagnostic Investigation	10 – 15 Minutes (SSH log digging)	4 Seconds (Non-destructive)	99.5% Faster Diagnostics
Runbook Maintenance	Stale wikis / 30% coverage	100% Dynamic & Self-Learning	Zero Documentation Debt
Remediation Verification	Exit code 0 assumption	In-Guest Socket & HTTP Proof	Zero False Positives
Engineer Workload	Severe alert fatigue & firefighting	75%+ Routine tasks offloaded	Focus on Platform Innovation

9. Conclusion & Strategic Engineering Roadmap

The Enterprise Autonomous ITSM Platform & Agentic SRE Control Tower demonstrates that enterprise AI in mission-critical infrastructure does not require sacrificing human oversight or system stability. By combining strict architectural decoupling, dynamic ReAct loops, hybrid vector search, in-guest verification, and single-click HITL governance, the platform delivers a production-ready blueprint for the future of autonomous Site Reliability Engineering.

Future Engineering Horizons: Upcoming platform milestones include eBPF kernel telemetry integration for sub-second socket anomaly detection, proactive incident prevention before user impact, and federated cross-cluster knowledge sharing across hybrid multi-cloud topologies.